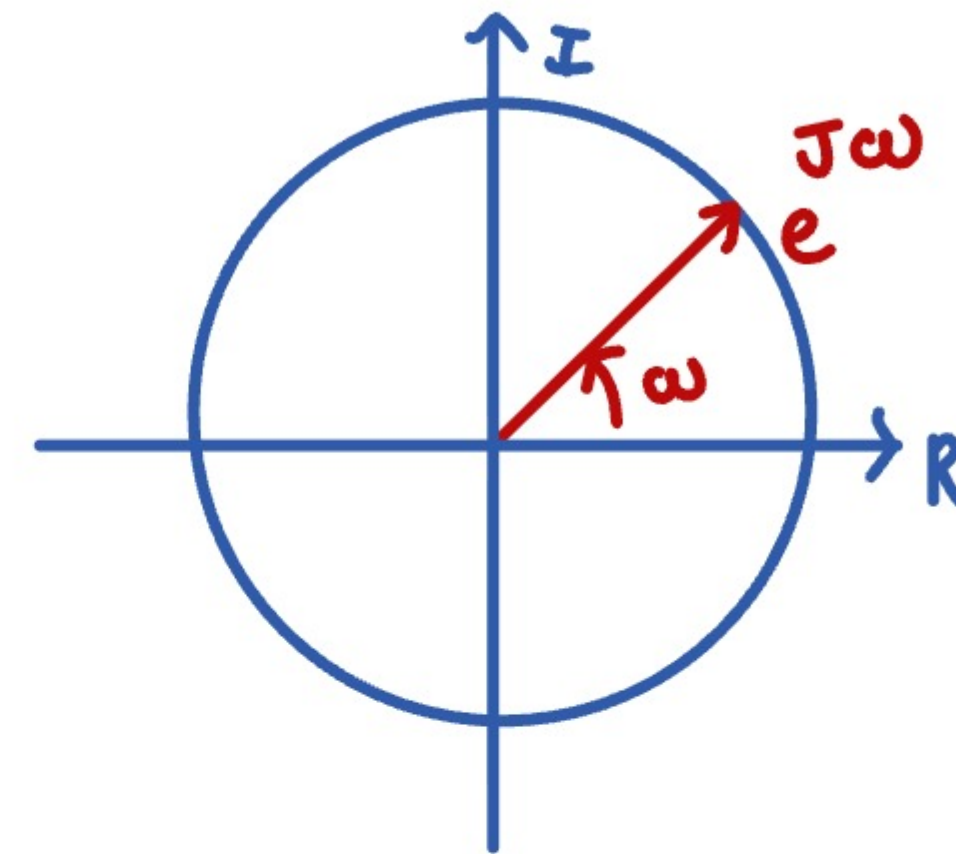


Tìm $x[n]$ c' ph' $X(\omega) = \begin{cases} 1, & |\omega| < \omega_c \\ 0, & \omega_c < \omega < \pi \end{cases}$



$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) \cdot e^{j\omega n} \cdot d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) \cdot e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} 1 \cdot e^{j\omega n} d\omega = \frac{1}{2\pi j n} \cdot e^{j\omega n} \Big|_{-\omega_c}^{\omega_c}$$

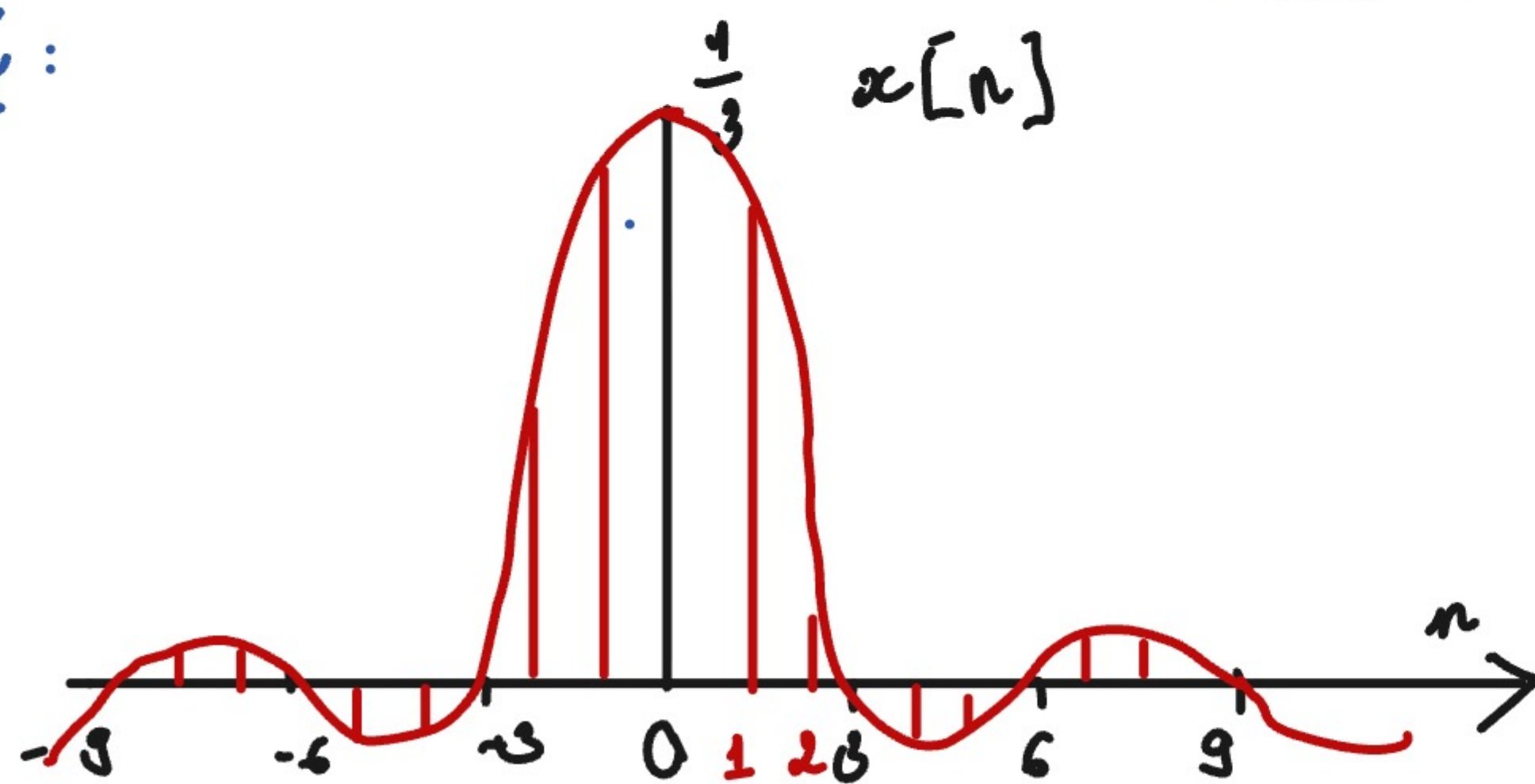
$$= \frac{1}{2\pi j n} \cdot [e^{j\omega_c n} - e^{-j\omega_c n}] = \frac{1}{2\pi j n} \cdot 2j \sin(\omega_c n) = \frac{\sin(\omega_c n)}{\pi n} = \frac{\sin(\pi \cdot \frac{\omega_c n}{\pi})}{\frac{\omega_c n}{\pi}} \cdot \frac{\omega_c}{\pi}$$

$$= \text{sinc}\left(\frac{\omega_c n}{\pi}\right) \cdot \frac{\omega_c}{\pi}$$

Cho' si' $\omega_c = \frac{\pi}{3}$, ta có: $F_c = f_c \cdot F_s = \frac{\omega_c}{2\pi} \cdot F_s = \frac{F_s}{6}$

$$x[n] = \frac{1}{3} \text{sinc}\left(\frac{n}{3}\right) = \begin{cases} \frac{1}{3} & (n=0) \\ 0 & (n=\pm 3, \pm 6, \pm 9) \\ x[n] \rightarrow 0 \text{ khi } n \rightarrow \pm\infty \end{cases}$$

Đồ thị:



Nhận xét: TH là xung CN tuần hoàn $\xleftrightarrow{\text{CTFS}}$ Ph' là hàm $\text{sinc}(\omega)$
Ph' là xung CN tuần hoàn $\xleftrightarrow{\text{DTFT}}$ TH là hàm $\text{sinc}(n)$